

Goutham Mukunda

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PROFESSIONAL SUMMARY

Curious and analytical Data Analyst with around 3 years of hands-on experience working with data pipelines, dashboards, and machine learning models. I enjoy turning messy data into clear insights that drive decisions. Skilled in Python, SQL, Tableau, and cloud tools like AWS, with a background in backend development that helps me understand how data flows end to end. I've worked on everything from building ETL processes to training models and presenting findings to stakeholders

EXPERIENCE

Data Analyst

Aug. 2024 – Present

Ford Motor Company

Remote, USA

- Implemented machine learning models to improve predictive analytics for vehicle diagnostics, reducing maintenance costs by 15% through anomaly detection in sensor data
- Analyzed over 1 million rows of vehicle telemetry data to uncover trends and detect anomalies, leading to a drop in maintenance issues using predictive models
- Built ETL pipelines with Python and SQL to clean, transform, and load vehicle health data into PostgreSQL and AWS S3 for reporting and model training
- Performed data profiling, validation, and anomaly detection on real-time vehicle sensor data using SQL and Python
- Designed and maintained Alteryx workflows for cleansing and reconciling structured and semi-structured telemetry data
- Implemented Informatica DQ rules for data quality checks on vehicle diagnostics feeds and integrated results with reporting pipelines
- Designed Tableau dashboards that visualized key metrics like engine performance, failure patterns, and diagnostic flags for engineering teams
- Collaborated with ML engineers to prepare time-series datasets and tune models using Scikit-learn and TensorFlow
- Wrote and optimized complex SQL queries to enable faster insights from live telemetry feeds, improving analysis time by 30%
- Helped automate daily data validation checks and reporting workflows, reducing manual effort across teams

Graduate Student Assistant

May. 2023 – Dec. 2023

University of Central Missouri

Lee's Summit, MO

- Assisted faculty with data preprocessing, feature selection, and model evaluation for research projects using Python, Scikit-learn, and Pandas
- Built internal tools and APIs using Flask to support data analysis workflows and expose ML models as services
- Maintained PostgreSQL databases and developed queries for research datasets involving student performance and behavioral analytics
- Conducted workshops for students on using Tableau, Power BI, and Excel to build dashboards and interpret data trends
- Supported classroom activities and guided students on hands-on labs involving Python for data cleaning and visualization

Software Developer

Jun. 2022 – Dec. 2022

Tvast IT Solutions

Bengaluru, India

- Built and maintained APIs using Django to support an inventory management system, improving real-time data tracking and reporting by 30%
- Wrote SQL queries and optimized data fetching logic across MySQL to reduce dashboard load times for business stakeholders
- Integrated OpenCV and PyTorch for automating document classification and basic visual data processing in onboarding pipelines
- Created interactive UI features using React.js to allow filtering, exporting, and analyzing inventory logs in the frontend

- Contributed to setting up CI/CD workflows using Jenkins and GitLab CI for automating testing and deployment of analytics tools
- Helped implement Kafka-based messaging for handling and visualizing real-time stock updates and alert notifications

Machine Learning Intern

Dec. 2021 – Jun. 2022

Mindtree LLC

Bengaluru, India

- Built a predictive analytics model using Python, Scikit-learn, and TensorFlow to identify potential customer churn, improving accuracy by 15% over baseline
- Engineered features and conducted hyperparameter tuning to improve model performance on classification tasks
- Processed and cleaned over 250,000 customer records using Pandas, significantly reducing preprocessing time and errors
- Created visualizations in Tableau and Matplotlib to communicate model outputs, customer segments, and key patterns to internal stakeholders
- Supported team in running A/B testing on ML predictions and translating results into actionable insights

PROJECTS

InsightExtractor: AI-Powered Article Analyzer

May 2024 – June 2024

- Developed an advanced tool using Streamlit, OpenAI's LLMs, LangChain, RAG techniques, and FAISS to analyze and extract insights from articles, allowing users to input URLs, process content, and retrieve detailed answers with sources. Achieved a 90% accuracy rate in information retrieval
- Implemented text splitting, vector embeddings, and robust query handling, ensuring real-time processing and precise information retrieval from the provided articles. Improved processing speed by 40%

Enterprise Data Quality Monitoring Platform

Mar. 2024 – Mar. 2024

- Designed a rule-based data quality monitoring solution using Informatica DQ and Alteryx, integrating with Snowflake and S3 data lakes
- Built an automated dashboard in Power BI to track rule violations, data completeness, and validity across critical business domains
- Integrated Collibra Catalog to attach quality scores and business definitions to datasets, enabling better data stewardship

LSTM for Time Series Prediction

Nov. 2023 – Jan. 2024

- Built an LSTM-based forecasting model to predict airline passenger trends using historical flight data
- Processed 100,000+ records using Pandas and NumPy, reducing data preprocessing time by 30%
- Deployed the model on AWS Lambda, enabling real-time demand prediction for airlines

Hand Gesture Recognition for Differently Abled People

Oct. 2021 – Apr. 2022

- Led award-winning (2nd Place, Project Exhibition-2022) deep learning project using Python, TensorFlow, Mediapipe, and OpenCV. Achieved 98% accuracy in recognizing hand gestures and translating them into voice/text, enhancing accessibility for speech-impaired users

TECHNICAL SKILLS

Languages: SQL, Python, Java

Data Quality & Governance: Collibra (DQ, Catalog), Unity Catalog, Azure Purview, Informatica, Talend, Alteryx, Metadata Management, Lineage Tracking, Business Glossary

Reporting Tools: Power BI, Tableau, Excel

Data Platforms: Snowflake, AWS S3, Redshift, Azure Synapse, Delta Lake, PostgreSQL, MySQL

ETL & Workflow: PySpark, Jenkins, Kafka, REST APIs, AWS Glue

Others: Flask, Git, Docker, Jupyter, Agile, Jira, Confluence

EDUCATION

University of Central Missouri

Lee's Summit, MO

Master of Science in Computer Science

Jan. 2023 – May 2024

Visvesvaraya Technological University

Bengaluru, India

Bachelor of Engineering in Computer Science and Engineering

Aug. 2018 – June 2022

ACHIEVEMENTS

Winner of Smart India Hackathon, a prestigious competition conducted by the Department of Education, Government of India

Runner-Up, HackFest-2021, conducted by ATME College of Engineering

Published Research: Hand Gesture Recognition for Differently Abled People, published in the Journal of Emerging Technologies and Innovative Research (JETIR), May 2022 <https://www.jetir.org/view?paper=JETIR2204140>